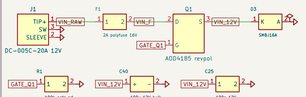
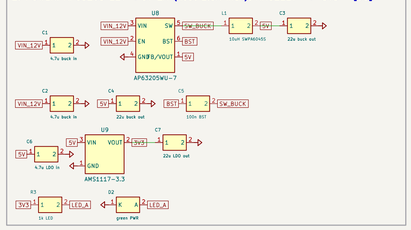
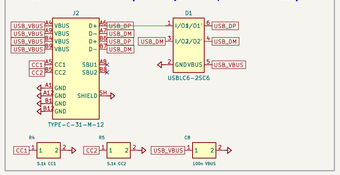
1. 12V INPUT: barrel -> 2A polyfuse -> P-FET revpol -> SMBJ16A TVS [ADR-0001]



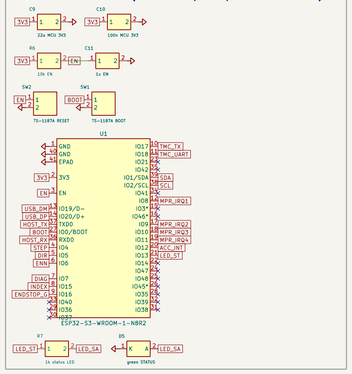
2. POWER: AP63205 12V->5V 2A (host 1.5A max); AMS1117 -> 3V3 [D5]



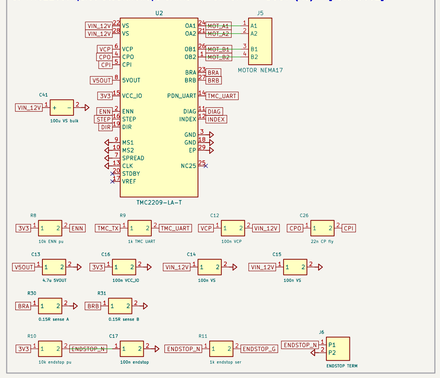
3. USB-C PROGRAMMING (data only; board powers from 12V)



4. ESP32-S3-WROOM-1 (native USB; pin map ARCHITECTURE.md)



5. TMC2209: quiet StealthChop, UART, MOTOR OFF AT BOOT (R8) [ADR-0002]



6. HOST HEADER: 5V/1.5A MAX + UART; I2C bus pullups [D8]

7. LIS2DH12 ACCEL: lid angle >= 20deg rule [D4]

8. CAPACITIVE: 4x MPR121QR2, 24 ELECTRODE LINES (short stubs!) [D4]

Cat toilet lid controller: ESP32-S3 + TMC2209 + 4x MPR121 + LIS2DH12; 01_docs/ + ADRs in repo

Sheet: /	
File: shitty_kitty.kicad_sch	
Title: shitty-kitty controller	
Size: A1 Date: 2026-07-18	**Rev: dev**
KiCad E.D.A. 10.0.4	Id: 1/1